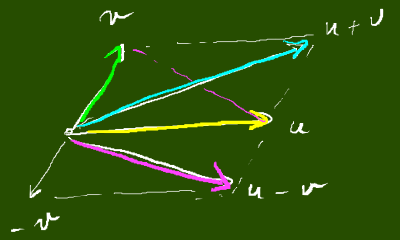
$\mathbb{R}^2$, $\mathbb{R}^3$

$V = \{ f : [a,b] \to \mathbb{R} \mid f \text{ cont} \}$

$\|f\| = \max_{x \in [a,b]} |f(x)|$

$f_n \xrightarrow{uc} f \iff \|f_n - f\| \to 0$



$\|u+v\|^2 + \|u-v\|^2 =$

$\langle u+v, u+v \rangle + \langle u-v, u-v \rangle =$

$\langle u,u \rangle + \langle u,v \rangle + \langle v,u \rangle + \langle v,v \rangle +$

$\langle u,u \rangle - \langle u,v \rangle - \langle v,u \rangle + \langle v,v \rangle$

$= 2 (\|u\|^2 + \|v\|^2)$

$\|u+v\|^2 = \|u\|^2 + 2\langle u,v \rangle + \|v\|^2$

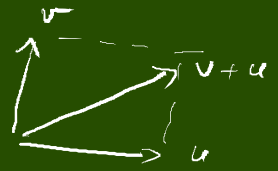
$\leq \|u\|^2 + 2|\langle u,v \rangle| + \|v\|^2$

$$\leq \|u\|^2 + 2\|u\|\cdot\|v\| + \|v\|^2$$

$$= (\|u\| + \|v\|)^2$$

$u, v \in V$ <u>ortogonali</u> $\iff \langle u, v \rangle = 0$

Pitagora : $\langle u, v \rangle = 0 \iff \|u+v\|^2 = \|u\|^2 + \|v\|^2$



$u, v \neq 0$

<u>$\cos\theta = 0$</u> $\iff \langle u, v \rangle = \|u\|\cdot\|v\|$

$\iff$? tesi

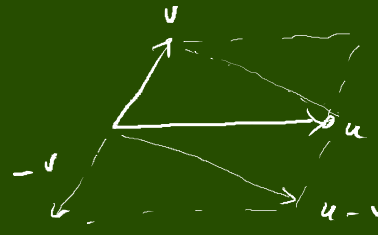
$\exists \lambda \cdot v = \lambda u$

$\lambda > 0$



$\cos\theta = \pi \iff \exists \lambda < 0 \cdot v = \lambda u$ tesi





X, $d : X \times X \longrightarrow \mathbb{R}_+$

- $d(u,v) = d(v,u)$ $\quad u, v \in X$
- $d(u,v) = 0 \Leftrightarrow u = v$
- нер. Δ.

(X, d) — *spațiu metric*

exemple 1) $d(10100111, 11010110) = 4$

10100111
11010110

HAMMING

2) 11010 → 11010

LEVENSHTEIN

$d(1010, 11010) = 1$

3)

E R T

S D F G

X C

---

1) $e_1 = \begin{pmatrix} 1 \\ 0 \\ \vdots \\ 0 \end{pmatrix}$ $e_2 = \begin{pmatrix} 0 \\ 1 \\ \vdots \\ 0 \end{pmatrix}$ $\langle e_1, e_2 \rangle =$

$1 \cdot 0 + 0 \cdot 1 + 0 \cdot 0 + \dots + 0 \cdot 0 = 0$

$\langle f, g \rangle = \int_0^{2\pi} f(x) g(x) \, dx$

$\cos 0x$

$\|\sin nx\| = ?$ $\|\cos nx\| = ?$

2) $1, \sin x, \cos x, \sin 2x, \cos 2x, \sin 3x, \cos 3x, \dots, \sin nx,$ lin. indep. $\cos nx$

$\langle \sin nx, \sin mx \rangle = 0$, $m \neq n$

$\langle \sin nx, \cos mx \rangle = 0$ $\forall\, m, n$

$\langle \cos nx, \cos mx \rangle = 0$ $\forall\, m \neq n$

$\langle 1, - \rangle \;|\; \lambda_0 \cdot 1 + \lambda_1 \cdot \cos x + \mu_1 \cdot \sin x + \lambda_2 \cdot \cos 2x + \dots = 0$

$\lambda_0 \underbrace{\langle 1, 1 \rangle}_{\neq 0} + \lambda_1 \underbrace{\langle 1, \cos x \rangle}_{0} + \mu_1 \underbrace{\langle 1, \sin x \rangle}_{0} + \dots = 0$

$\Rightarrow \lambda_0 = 0$

$\langle \varphi x, - \rangle \Big| \lambda_0 \cdot \underbrace{\langle \varphi x, 1 \rangle}_{=0} + \lambda_1 \underbrace{\langle \varphi x, \varphi x \rangle}_{\neq 0} + \dots = 0$

$\Rightarrow \lambda_1 = 0$.

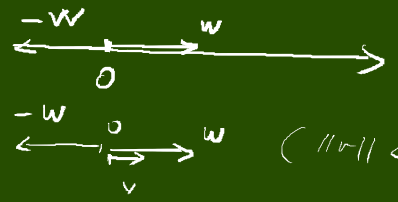
$v \in V, \; v \neq 0 \quad \Rightarrow \quad \exists \; w \in Sp\{v\}$

$\|w\| = 1$

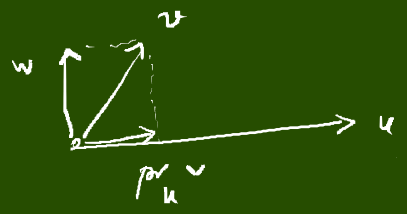
$w = \frac{1}{\|v\|} v$

$\|w\| = \left\| \frac{1}{\|v\|} v \right\| = \left| \frac{1}{\|v\|} \right| \cdot \|v\| = 1$

$\| -w \| = |-1| \cdot \|w\|$



$\mathbb{R}^2, \mathbb{R}^3$



$w = v - pr_u v$ $\begin{cases} w \perp u \\ pr_u v \in Sp\{u\} \end{cases}$

$(V, \langle , \rangle)$ $\quad u, v \in V$, $u \neq 0$

Unicitatea

$v = w_1 + w_2 = z_1 + z_2$

$\begin{cases} w_1 \in Sp\{u\} \\ \langle w_2, u \rangle = 0 \end{cases}$ $\quad \begin{cases} z_1 \in Sp\{u\} \\ \langle z_2, u \rangle = 0 \end{cases}$

$\Rightarrow$ $\underbrace{w_1 - z_1}_{\in Sp\{u\}} = z_2 - w_2$

$w_1 - z_1 = \alpha \cdot u$, $\alpha \in \mathbb{R}$

$\Rightarrow \langle z_2 - w_2, z_2 - w_2 \rangle = \langle z_2 - w_2, w_1 - z_1 \rangle =$

$$= \langle z_2 - w_2, \alpha u \rangle = \alpha \left(\underbrace{\langle z_2, u \rangle}_{0} + \underbrace{\langle w_2, u \rangle}_{0} \right) = 0$$

$$\langle = \rangle \quad z_2 - w_2 = 0 \quad \Longleftrightarrow \quad z_2 = w_2 \Rightarrow z_1 = w_1$$

Existență caut $w_1 \in \mathrm{Sp}\{u\}$ a.î. $v = w_1 + w_2$

$$\langle w_2, u \rangle = 0$$

$w_1 \in \mathrm{Sp}\{u\} \Rightarrow \exists \alpha \in \mathbb{R} \quad w_1 = \alpha \cdot u$

$v = \alpha u + w_2$

$\langle w_2, u \rangle = 0$

$\langle v - \alpha u, u \rangle = 0 \iff \langle v, u \rangle - \alpha \langle u, u \rangle = 0$

$$\Rightarrow \quad \alpha = \frac{\langle v, u \rangle}{\langle u, u \rangle} \in \mathbb{R}$$

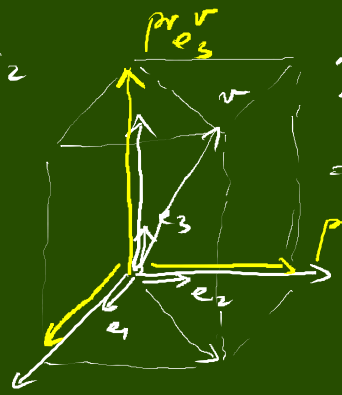
$\langle u, u \rangle \neq 0$ (pt. că $u \neq 0$)

- $w_1 = \alpha u = \dfrac{\langle v, u \rangle}{\langle u, u \rangle} u \in \mathrm{Sp}\{u\}$

$\langle u, u \rangle = \|u\|^2$

- $w_2 = v - w_1$ $\perp u$ $(\langle w_2, u \rangle = 0)$

$v = w_1 + w_2$



$v = \text{pr}_{e_1} v + \text{pr}_{e_2} v + \text{pr}_{e_3} v$

$= \sum_{i=1}^{3} \frac{\langle v, e_i \rangle}{\langle e_i, e_i \rangle} e_i$

Exemplu.

$v = \begin{pmatrix} 3 \\ 2 \\ 1 \end{pmatrix}$ $e_1 = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}$

$\text{pr}_{e_1} v = ? \begin{pmatrix} 3 \\ 0 \\ 0 \end{pmatrix} = 3e_1$

$= \frac{\langle v, e_1 \rangle}{\langle e_1, e_1 \rangle} e_1 = \frac{3 \cdot 1 + 2 \cdot 0 + 0 \cdot 0}{1 \cdot 1 + 0 \cdot 0 + 0 \cdot 0} e_1$

$= 3e_1$

$w_1 \in \text{Sp}\{u\}$

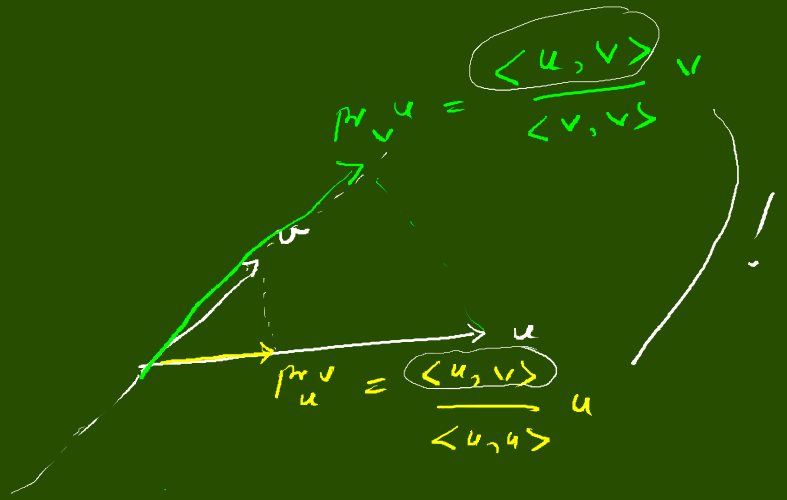
ortogonali

$d(v, w)^2 = \|v - w_1\|^2 = \|\underbrace{(v - \text{pr}_u v)}_{\perp u} + \underbrace{(\text{pr}_u v - w_1)}_{\in \text{Sp}\{u\}}\|^2 \overset{\text{Pitagora}}{=}$

$$= \|v - \mathrm{pr}_u v\|^2 + \underbrace{\|\mathrm{pr}_u v - w\|^2}_{\geq 0} \geq$$

$$\geq \|v - \mathrm{pr}_u v\|^2$$

$$\Rightarrow \|v - w\| \geq \|v - \underbrace{\mathrm{pr}_u v}_{\in \mathrm{Sp}\, u}\| \qquad \forall\, w \in \mathrm{Sp}\{u\}$$



$(V, \langle , \rangle)$ base ortogonalo $\{v_1, \ldots, v_n\}$

$v \in V \Rightarrow \exists! \lambda_1 \dots \lambda_n \in \mathbb{R}$ a.i.

$\langle -, v_i \rangle \mid v = \lambda_1 v_1 + \dots + \lambda_n v_n \qquad \lambda_1 \dots \lambda_n = ?$

$\langle v, v_1 \rangle = \langle \lambda_1 v_1, v_1 \rangle + \langle \lambda_2 v_2, v_1 \rangle + \dots + \langle \lambda_n v_n, v_1 \rangle$

$= \lambda_1 \langle v_1, v_1 \rangle + \lambda_2 \underbrace{\langle v_2, v_1 \rangle}_{0} + \dots + \lambda_n \underbrace{\langle v_n, v_1 \rangle}_{0}$

$\Rightarrow \lambda_1 = \dfrac{\langle v, v_1 \rangle}{\langle v_1, v_1 \rangle}$ analog $\lambda_i = \dfrac{\langle v, v_i \rangle}{\langle v_i, v_i \rangle}$, $i \in \overline{1,n}$

$v = \lambda_1 v_1 + \dots + \lambda_n v_n$

$= \dfrac{\langle v, v_1 \rangle}{\langle v_1, v_1 \rangle} v_1 + \dots$

proiecția lui v pe v_1

$\langle u, v \rangle = \langle x_1 v_1 + \dots + x_n v_n, y_1 v_1 + \dots + y_n v_n \rangle =$

$= x_1 y_1 \underbrace{\langle v_1, v_1 \rangle}_{1} + x_1 y_2 \underbrace{\langle v_1, v_2 \rangle}_{0} + \dots + x_1 y_n \underbrace{\langle v_1, v_n \rangle}_{0}$

$$+ \dots + x_n y_n \underbrace{\langle v_n, v_n \rangle}_{1}$$

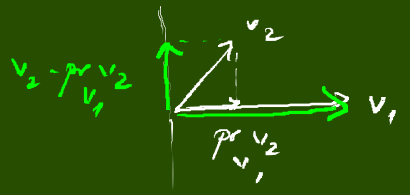
$$= \underbrace{x_1 y_1 + \dots + x_n y_n}$$

$$V \longrightarrow \mathbb{R}^n$$

$$=$$

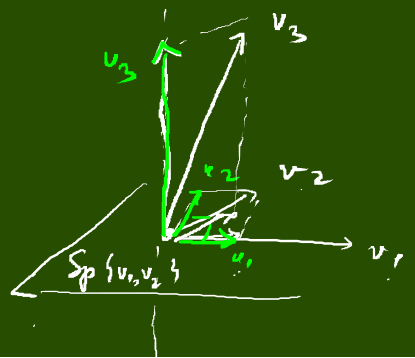
$n = 2$

$\{v_1, v_2\}$ — bază oarecare $\longrightarrow$ $\{u_1, u_2\}$ — bază ortogonală $\longrightarrow$ $\{e_1, e_2\}$ — bază ortonormată



$$\begin{cases} u_1 = v_1 \\ u_2 = v_2 - \text{pr}_{u_1} v_2 \end{cases}$$

$$\begin{cases} e_1 = \frac{1}{\|u_1\|} u_1 \\ e_2 = \frac{1}{\|u_2\|} u_2 \end{cases}$$



$$\begin{cases} u_1 = v_1 \\ u_2 = v_2 - \text{pr}_{u_1} v_2 \\ u_3 = v_3 - \text{pr}_{u_1} v_3 - \text{pr}_{u_2} v_3 \end{cases}$$

$$\langle u_3, u_1 \rangle = 0$$

$$\langle v_3 - \operatorname{pr}_{u_1} v_3 - \operatorname{pr}_{u_2} v_3, u_1 \rangle =$$

$$= \langle v_3, u_1 \rangle - \langle \operatorname{pr}_{u_1} v_3, u_1 \rangle - \underbrace{\langle \operatorname{pr}_{u_2} v_3, u_1 \rangle}_{0}$$

$$= \langle v_3, v_1 \rangle - \frac{\langle v_3, u_1 \rangle}{\langle u_1, u_1 \rangle} \langle u_1, u_1 \rangle = 0$$